

Instructions

This mock mirrors the BI3 selection flow: aptitude + reasoning, then data-engineering technical MCQs, then SQL and coding.

Section 1 - Aptitude (6 Q)

Section 2 - Reasoning (5 Q)

Section 3 - Data Engineering Technical (8 Q)

Section 4 - SQL (2 problems)

Section 5 - Coding / Python (2 problems)

Section 1 - Aptitude

Q1. What is 25% of 240?

(A) 50

(B) 60 ✓

(C) 70

(D) 80

Answer: (B) 60

Q2. A shirt costs Rs. 800 and is sold at a 10% discount. The selling price is:

(A) Rs. 700

(B) Rs. 720 ✓

(C) Rs. 750

(D) Rs. 680

Answer: (B) Rs. 720

Q3. If 3 pens cost Rs. 45, how much do 7 pens cost?

(A) Rs. 90

(B) Rs. 105 ✓

(C) Rs. 100

(D) Rs. 120

Answer: (B) Rs. 105

Q4. The average of 12, 18 and 24 is:

(A) 16

(B) 18 ✓

(C) 20

(D) 22

Answer: (B) 18

Q5. Divide Rs. 500 in the ratio 2 : 3. The larger share is:

(A) Rs. 200

(B) Rs. 300 ✓

(C) Rs. 250

(D) Rs. 350

Answer: (B) Rs. 300

Q6. A can finish a task in 12 days and B in 6 days. Together they finish in:

(A) 3 days

(B) 4 days ✓

(C) 5 days

(D) 6 days

Answer: (B) 4 days

Section 2 - Reasoning

Q1. Find the next term: 3, 6, 9, 12, ?

(A) 14

(B) 15 ✓

(C) 16

(D) 18

Answer: (B) 15

Q2. Find the next term: 2, 4, 8, 16, ?

(A) 24

(B) 30

(C) 32 ✓

(D) 20

Answer: (C) 32

Q3. Find the next term: 5, 10, 17, 26, ?

(A) 35

(B) 37 ✓

(C) 36

(D) 40

Answer: (B) 37

Q4. If RED is coded as 27 ($18+5+4$), then BLUE ($2+12+21+5$) is coded as:

(A) 38

(B) 40 ✓

(C) 42

(D) 36

Answer: (B) 40

Q5. If in a code CAT = DBU (each letter shifted by +1), then DOG is:

(A) EPH ✓

(B) EPG

(C) FPH

(D) DPG

Answer: (A) EPH

Section 3 - Data Engineering Technical

Q1. Which SQL clause is used to filter rows AFTER aggregation (on grouped results)?

(A) WHERE

(B) HAVING ✓

(C) GROUP BY

(D) ORDER BY

Answer: (B) HAVING

Q2. Which JOIN returns all rows from the left table and matching rows from the right, with NULLs where there is no match?

(A) INNER JOIN

(B) LEFT JOIN ✓

(C) CROSS JOIN

(D) SELF JOIN

Answer: (B) LEFT JOIN

Q3. A PRIMARY KEY constraint ensures that a column's values are:

(A) Sorted

(B) Unique and not null ✓

(C) Indexed only

(D) Foreign

Answer: (B) Unique and not null

Q4. In dimensional modeling, the central table containing measures/metrics is called the:

(A) Dimension table

(B) Fact table ✓

(C) Lookup table

(D) Bridge table

Answer: (B) Fact table

Q5. A schema with a central fact table connected directly to denormalized dimension tables is called a:

(A) Snowflake schema

(B) Star schema ✓

(C) Galaxy schema

(D) Network schema

Answer: (B) Star schema

Q6. A snowflake schema differs from a star schema mainly because its dimension tables are:

(A) Denormalized

(B) Normalized into sub-dimensions ✓

(C) Removed

(D) Combined

Answer: (B) Normalized into sub-dimensions

Q7. In the acronym ETL, the letters stand for:

(A) Extract, Transform, Load ✓

(B) Export, Transfer, Link

(C) Extract, Transfer, Log

(D) Evaluate, Transform, Load

Answer: (A) Extract, Transform, Load

Q8. The key difference between ETL and ELT is that in ELT:

(A) Data is never transformed

(B) Transformation happens after loading into the target system ✓

(C) Data is only extracted

(D) Loading is skipped

Answer: (B) Transformation happens after loading into the target system

Section 4 - SQL

Problem 1: Second Highest Salary

Find the second highest distinct salary from the employee table.

```
SELECT MAX(salary) AS second_highest
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee);
```

Expected Output

```
second_highest
42000
```

Problem 2: Nth Highest Salary using LIMIT/OFFSET

Find the 3rd highest distinct salary using ORDER BY with LIMIT and OFFSET.

```
SELECT DISTINCT salary
FROM emp
ORDER BY salary DESC
LIMIT 1 OFFSET 2;
```

Expected Output

```
salary
200
```

Section 5 - Coding / Python

Problem 1: Two Sum

Difficulty: *Easy* | Pattern: *Hash Map*

Problem Statement

Given an array of integers and a target value, return the indices of the two numbers that add up to the target.

Input Format

Line 1: the array. Line 2: the target.

Output Format

The two indices, space-separated.

Constraints

$2 \leq n \leq 10^4$; exactly one valid answer exists.

Sample Input

```
3 2 4
6
```

Sample Output

```
1 2
```

Python Solution

```
def two_sum(nums, target):
    seen = {}
    for i, x in enumerate(nums):
        if target - x in seen:
            return (seen[target - x], i)
        seen[x] = i

i, j = two_sum([3, 2, 4], 6)
print(i, j)
```

Step-by-Step Explanation

Store each number's index while scanning. If the complement (target - x) was already seen, return the stored index and the current index. $2 + 4 = 6$.

Dry Run

```
i=0 x=3 need=3 not seen; store 3->0
i=1 x=2 need=4 not seen; store 2->1
i=2 x=4 need=2 seen at 1 -> (1,2)
```

Time Complexity: **$O(N)$** | Space Complexity: **$O(N)$**

Problem 2: Happy Number

Difficulty: *Easy* | Pattern: *Cycle Detection (Hash Set)*

Problem Statement

A happy number is defined by repeatedly replacing the number with the sum of the squares of its digits; if this process reaches 1 the number is happy, otherwise it loops endlessly. Determine whether n is a happy number.

Input Format

A single integer n.

Output Format

'True' if n is happy, otherwise 'False'.

Constraints

$1 \leq n \leq 2^{31} - 1$

Sample Input

19

Sample Output

True

Python Solution

```
def is_happy(n):
    seen = set()
    while n != 1 and n not in seen:
        seen.add(n)
        n = sum(int(d) ** 2 for d in str(n))
    return n == 1

print(is_happy(19))
```

Step-by-Step Explanation

Repeatedly transform n into the sum of the squares of its digits. Track seen values to detect an endless loop. If we reach 1 the number is happy. 19 -> 82 -> 68 -> 100 -> 1.

Dry Run

```
19 -> 1^2+9^2 = 82
82 -> 6^2+2^2 = 68
68 -> 3^2+6^2 = 100
100 -> 1 -> stop, n==1 -> True
```

Time Complexity: **$O(\log n)$ per step** | Space Complexity: **$O(\log n)$**

BI3 Technologies

Full-Length Mock Test

Applied Role(s):

Data Engineer - Trainee

Online Assessment Preparation Material

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Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

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